

Reliability-Aware Survival Machine Learning for Credit Default and Prepayment Risk in Fintech

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Abstract

Credit default prediction is usually framed as a binary classification problem, but this framing ignores the timing of default and discards information from censored loans that are still active at the observation cutoff. Survival analysis offers a more natural formulation by treating default and prepayment as time-to-event processes with explicit handling of censoring. This paper proposes applying reliability-theoretic structure—specifically cohort-level hazard ordering and mean residual lifetime estimation—as additional modeling constraints within machine learning survival models for fintech credit risk. The core question is whether incorporating such constraints improves calibration, temporal stability, and cohort-level consistency compared with otherwise equivalent unconstrained survival models. A concrete experimental plan is outlined using publicly available Freddie Mac mortgage data, covering baseline comparisons, ablation studies, and evaluation metrics. This is an early-stage research proposal; no experimental results are reported here.

Keywords: credit default, prepayment risk, survival analysis, hazard function, reliability theory, machine learning, censoring, calibration, competing risks, fintech.

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1. Introduction

When a lending institution extends credit, it is exposed to two primary risks over the life of a loan: the borrower may default, failing to meet payment obligations, or the borrower may prepay, retiring the debt earlier than scheduled. Both events have direct financial consequences and neither is well-captured by a model that simply asks whether an event will occur. The timing of the event matters. A default in month three of a thirty-year mortgage is a very different outcome from a default in year twenty, even if both are labeled identically in a binary prediction framework.

The standard machine learning approach to credit risk—training a classifier to output a default probability—is useful, but it makes a significant simplification. It collapses a trajectory into a single number, ignores the fact that many loans in the training data are still active (and therefore have an unobserved outcome), and provides no natural way to answer horizon-specific questions such as: what is the probability that this loan will default within the next twelve months? What is the expected remaining time before default?

Survival analysis addresses these gaps directly. It is the branch of statistics concerned with modeling the time until an event occurs, with explicit treatment of censoring and the option to estimate event-time distributions rather than just event indicators. Reliability theory, developed largely in engineering contexts, deals with the same class of problems but brings additional conceptual tools: hazard ordering, stochastic ordering, remaining-lifetime analysis, and models for heterogeneous populations. These ideas translate naturally to credit risk.

This paper is an early-stage research proposal written as part of a first-year BS-MS project at IISER Tirupati. The aim is to define a testable research direction, not to report completed results. The core idea—that reliability-theoretic concepts such as hazard ordering and mean residual lifetime can be operationalized as regularization terms in ML survival models for credit risk—is, to the best of my knowledge, not well explored in the applied fintech literature. Whether the approach actually improves model performance is an empirical question, and the experimental plan described in later sections is designed to test it honestly, including the possibility that the proposed method does not clearly outperform simpler baselines. The paper reviews the relevant background in survival analysis and reliability theory, describes the proposed modeling ideas, and outlines a structured experimental plan using publicly available mortgage data.

2. Credit Risk as a Time-to-Event Problem

2.1. Inadequacy of Binary Classification

Consider a training dataset of mortgage loans. For a loan originated in 2010 with a 30-year term, we might observe monthly payment data from 2010 to 2023. Some loans will have defaulted; others will have been prepaid; many will simply be ongoing. A binary classification model applied to this dataset must decide what to do with the ongoing loans: it can either drop them (losing information), treat them as non-defaults (introducing a systematic error), or apply ad hoc procedures.

Beyond the censoring problem, binary classification cannot directly express uncertainty that is time-dependent. A model might correctly estimate that a loan has a 10% default probability over its full lifetime, but this estimate is not useful if the analyst needs a 12-month probability, a 24-month probability, and an expected time to default simultaneously.

2.2. Defining Events

For credit modeling, the following event definitions are standard. *Credit default* occurs when a loan reaches a state of serious delinquency or enters foreclosure or loss proceedings, as defined in the dataset. *Prepayment* occurs when a loan is voluntarily paid off before its scheduled maturity. *Censoring* applies to a loan that is still active at the data cutoff and has experienced neither default nor prepayment. A censored observation is not a non-event; it is an observation whose event time is known only to exceed the observation window.

2.3. Competing Risks

Default and prepayment are not independent. Once a loan prepays, it cannot subsequently default within its original lifecycle. These events therefore constitute *competing risks*: the occurrence of one precludes the other [Fine and Gray, 1999]. Treating them as independent is an approximation that can distort probability estimates, particularly in environments where prepayment is frequent. A correct treatment requires either cause-specific hazard models or subdistribution hazard models, such as the approach of Fine and Gray [1999].

2.4. Formal Setup

Let $T \geq 0$ denote the time from loan origination (or first observation) to the event of interest, and let $C \geq 0$ denote a censoring time. The observed quantities are $\tilde{T} = \min(T, C)$ and the event indicator $\Delta = \mathbf{1}(T \leq C)$.

The *survival function* is

$$S(t) = P(T > t), \tag{1}$$

which gives the probability that the loan survives (has not defaulted or prepaid) beyond time t . The *hazard function* is

$$h(t) = \lim_{\delta \rightarrow 0} \frac{P(t \leq T < t + \delta \mid T \geq t)}{\delta}, \quad (2)$$

the instantaneous rate of event occurrence given survival to time t . The two quantities are related by $S(t) = \exp\left(-\int_0^t h(u) du\right)$. The cumulative hazard is $H(t) = \int_0^t h(u) du$.

In a competing-risks setting, separate cause-specific hazard functions $h_1(t)$ (default) and $h_2(t)$ (prepayment) are defined, and the overall survival function is $S(t) = \exp(-H_1(t) - H_2(t))$.

3. Survival Analysis and Reliability Theory

3.1. Classical Survival Models

The *Cox proportional hazards model* [Cox, 1972] is the most widely used regression model for survival data. It specifies:

$$h(t \mid \mathbf{x}) = h_0(t) \exp(\boldsymbol{\beta}^\top \mathbf{x}), \quad (3)$$

where $h_0(t)$ is an unspecified baseline hazard and $\mathbf{x}$ is a vector of covariates. The model assumes proportional hazards: the ratio of hazards between any two covariate profiles is constant over time. This is a strong assumption that may not hold in credit data, where the relative risk of different borrower groups can shift over the loan lifecycle.

Accelerated failure time (AFT) models take a different approach. Instead of scaling the hazard, they model the logarithm of event time as a linear function of covariates:

$$\log T = \boldsymbol{\beta}^\top \mathbf{x} + \sigma \epsilon, \quad (4)$$

where ϵ follows a specified distribution (normal, logistic, or extreme value) and σ is a scale parameter. Weibull AFT models are among the most common. AFT models are sometimes more interpretable in financial applications because the coefficient β_j directly measures how a unit increase in x_j multiplicatively accelerates or decelerates the expected time to event.

3.2. Reliability Theory Concepts

Reliability theory was developed to study the lifetime of engineered systems, but many of its concepts map well onto credit risk.

The *mean residual lifetime* (MRL) function, $m(t) = E(T - t \mid T > t)$, gives the expected remaining lifetime of a unit that has survived to time t . For an active loan, this corresponds

to the expected remaining time to default. MRL is a more natural quantity than the survival function alone for risk management, because it provides a direct measure of how much time-at-risk remains.

Stochastic ordering formalizes the notion that one distribution is uniformly riskier than another. A random variable T_A is said to be stochastically smaller than T_B (written $T_A \leq_{\text{st}} T_B$) if $S_A(t) \leq S_B(t)$ for all t , meaning borrower group A has shorter expected survival times than group B . The related concept of *hazard ordering* ($T_A \leq_{\text{hr}} T_B$) requires that $h_A(t) \geq h_B(t)$ for all t : group A faces higher instantaneous risk at every point in time [Shaked and Shanthikumar, 2007].

These ordering concepts are useful in credit risk for the following reason. We have strong empirical and economic prior beliefs about how borrower cohorts should order. Loans with high loan-to-value (LTV) ratios and low credit scores should, on average, have higher hazard than loans with low LTV and high credit scores. Severely delinquent loans should have higher near-term hazard than current loans. A model that violates these orderings—even if it has high average discriminatory power—is producing predictions that are difficult to explain and may reflect overfitting to spurious patterns in the training data.

Mixture models in reliability theory accommodate population heterogeneity by treating the observed population as a mixture of subgroups, each with its own lifetime distribution. In credit portfolios, borrowers are genuinely heterogeneous: some are structurally low-risk and likely to prepay early, others are moderate-risk and likely to survive a long time, and a minority are high-risk and prone to early default. A global single-population model may fit the aggregate data reasonably well but may produce poorly calibrated predictions at the individual level.

4. Machine Learning Methods for Survival Prediction

4.1. Motivation for ML Survival Models

Classical survival models have important limitations when applied to large-scale tabular credit data with complex covariate interactions. Cox models assume proportional hazards and handle nonlinearity poorly without manual feature engineering. Parametric AFT models impose distributional assumptions that may not hold. These limitations have motivated the development of machine learning approaches that use flexible function approximators while retaining the survival-analysis structure.

4.2. Random Survival Forests

[Ishwaran et al. \[2008\]](#) extended random forests to the survival setting. At each tree node, a log-rank splitting criterion is used to maximize separation of survival curves between child nodes. The ensemble aggregates cumulative hazard estimates (via the Nelson–Aalen estimator) across trees to produce a borrower-level survival curve. Random survival forests handle nonlinearity, interactions, and missing values naturally, and they remain competitive with more complex models on many tabular datasets.

4.3. Gradient-Boosted Survival Models

Gradient boosting can be adapted to survival objectives in at least two ways. One approach uses the negative partial log-likelihood of the Cox model as the loss function and fits an ensemble of regression trees to its gradient [[Ridgeway, 1999](#)]. Another approach, implemented in packages such as `XGBoost` and `LightGBM`, supports AFT objectives directly, fitting the parameters of a parametric survival distribution. Gradient boosting tends to perform well on tabular data with moderate-to-large sample sizes, which makes it a natural choice for mortgage loan datasets.

4.4. Deep Survival Models

[Katzman et al. \[2018\]](#) proposed DeepSurv, a neural network that estimates a proportional hazards model where the linear predictor is replaced by a neural network, allowing arbitrary nonlinear covariate effects. [Lee et al. \[2018\]](#) introduced DeepHit, which does not assume a specific hazard structure but instead directly estimates the probability mass function over a discretized time grid, making it compatible with competing risks. Deep models can in principle capture complex interactions and can be adapted to handle time-varying covariates. Their main practical disadvantages are greater computational cost, sensitivity to hyperparameters, and reduced interpretability compared to tree-based models.

4.5. Time-Varying Covariates

Credit data are naturally longitudinal. Monthly delinquency status, unpaid principal balance, and payment history evolve over time and contain strong predictive information about near-term default risk. Landmark models handle time-varying covariates by re-defining the prediction problem at a fixed lookback window [[van Houwelingen and Putter, 2011](#)]. Joint models and extended Cox models can also accommodate time-varying covariates directly, though at the cost of greater modeling complexity.

5. Reliability-Aware Modeling Ideas

The central proposal of this review is that reliability-theoretic concepts can be operationalized as modeling choices or regularization terms in machine learning survival models, improving calibration, interpretability, and temporal robustness. This section describes the main ideas.

5.1. Hazard-Ordering Regularization

Consider a partition of borrowers into cohorts A and B , where A is empirically riskier than B based on origination characteristics (e.g., A has LTV $> 90\%$ and credit score < 620 , while B has LTV $< 70\%$ and credit score > 720). Under hazard ordering, we expect $h_A(t) \geq h_B(t)$ for all t in the observation window.

A reliability-aware survival model can be trained with an additional penalty term that discourages predicted hazard violations:

$$\mathcal{L}_{\text{reg}} = \lambda \sum_{t \in \mathcal{T}} \max(0, \hat{h}_B(t) - \hat{h}_A(t)), \quad (5)$$

where $\mathcal{T}$ is a set of evaluation time points and $\lambda > 0$ controls the strength of the regularization. This penalty is zero when the ordering is satisfied and positive otherwise. The motivation is not to override the data, but to constrain the model to avoid predictions that violate well-established economic priors, particularly when training data is sparse for certain cohort-time combinations.

Whether such a regularization actually improves held-out performance is an empirical question that should be tested with appropriate ablations. It is plausible that the benefit is larger for small cohort sizes and for out-of-time prediction, where the model has less direct evidence.

5.2. Remaining-Lifetime Estimation for Active Loans

For portfolio management, a useful quantity is the expected remaining time to default for loans that are currently active. Given a borrower's current survival function $\hat{S}(t)$ estimated from a fitted model, the mean residual lifetime at the current loan age t_0 is

$$\hat{m}(t_0) = \frac{\int_{t_0}^{\infty} \hat{S}(u) du}{\hat{S}(t_0)}. \quad (6)$$

In practice, the integral is approximated over a finite horizon. This quantity is directly actionable: a risk team can rank active loans by estimated remaining lifetime, prioritize early intervention for loans with short estimated remaining time, and construct forward-

looking expected-loss estimates at the portfolio level.

5.3. Mixture Models for Borrower Heterogeneity

A latent mixture survival model assumes that the population consists of K unobserved subgroups, each with a distinct survival distribution:

$$S(t \mid \mathbf{x}) = \sum_{k=1}^K \pi_k(\mathbf{x}) S_k(t \mid \mathbf{x}), \quad (7)$$

where $\pi_k(\mathbf{x})$ is the (covariate-dependent) mixing weight for group k . Mixture models can be estimated by expectation-maximization or variational inference. The interpretation is natural, though the choice of K introduces a model-selection problem, and the labels assigned to latent groups should not be over-interpreted without external validation.

5.4. Copula-Based Portfolio Dependence

In a credit portfolio, default events are not independent across borrowers. Geographic concentration, servicer practices, and macroeconomic conditions can create correlated defaults. Copula models separate the marginal default distributions from the dependence structure, allowing the joint portfolio loss distribution to be estimated flexibly [Li, 2000]. Archimedean copulas (Clayton, Gumbel, Frank) are commonly used in credit applications because of their tractability and their ability to capture tail dependence. This extension is more technically involved and is best treated as a secondary modeling component after the individual-level survival model is well-specified.

6. Main Contribution and Experimental Goal

The central question of this project is narrow: does incorporating reliability-theoretic structure into a survival ML model produce measurable improvements in calibration, hazard-order consistency, and temporal stability compared with an otherwise equivalent unconstrained model of the same family? This is not a question about whether survival models beat logistic regression—that comparison provides useful context but is not the primary focus.

The proposed approach, which I call the Reliability-Regularized Survival Model (RRSM), adds a cohort-level hazard-ordering penalty to a standard gradient-boosted survival objective. The penalty discourages the model from predicting that lower-risk borrower cohorts face higher hazard than higher-risk ones, based on origination characteristics such as credit score and LTV ratio. In addition to the ordering penalty, the framework estimates mean residual lifetime for active loans, providing a forward-looking risk measure

that is more directly actionable than a single predicted probability.

Three properties are targeted for evaluation. First, *calibration*: predicted default probabilities at 12, 24, and 36-month horizons should agree with observed event rates in the corresponding population. Second, *hazard-order consistency*: cohort-level predicted hazards should conform to economically motivated orderings. Third, *temporal stability*: model performance should not degrade substantially when evaluated on later vintages not seen during training.

The primary comparison in the experiments is between the RRSM and an unconstrained survival model of the same base architecture. Classification baselines and classical survival models are included to establish that the survival framing is justified and to provide broader context. Whether the proposed regularization genuinely helps—and under what conditions—is an open empirical question that the experiments are designed to answer.

7. Dataset Construction and Event-Code Mapping

7.1. File Structure and Datasets

The Freddie Mac Single-Family Loan-Level Dataset provides two file types per origination group: an origination file containing static loan characteristics recorded at the time of loan acquisition, and a monthly performance file containing one record per loan per active reporting month [Freddie Mac, 2024]. The two files are joined on the loan sequence number. The Fannie Mae Single-Family Loan Performance Data has a comparable structure and is used for external validation [Fannie Mae, 2024].

7.2. Event-Code Mapping

Table 1 documents the mapping from dataset status codes to model event labels. Exact numeric codes must be verified against the official user guide [Freddie Mac, 2024] before use.

Table 1: Event-code mapping for the Freddie Mac dataset. Entries marked *TBV* are to be verified against the official data dictionary [Freddie Mac, 2024].

Event Label	Column	Code / Condition	Note
Default	Zero Balance Code	TBV (e.g., foreclosure, REO)	Terminal state
Prepayment	Zero Balance Code	TBV (e.g., payoff, repurchase)	Voluntary payoff
Censored	Zero Balance Code	Still active at cutoff	Right-censored
Censored (other)	Zero Balance Code	TBV (other exit reasons)	Treat as censored
Origination date	Origination Date column	YYYY-MM	Time origin
Terminal month	Last performance record	Month of exit	Event time (months)

7.3. Panel Construction

Starting from the raw performance file, sort all records by loan identifier and reporting month. Assign each loan a month index starting from one at the first observed performance record. Identify the first month in which a terminal status code (default or prepayment) appears; this month index is the observed event time. For loans that never reach a terminal status, the event time is the total number of observed months and the event indicator is set to censored. Remove records for months after the terminal event month.

7.4. Train, Validation, and Test Splits

A vintage-based split is used. A suggested configuration is: training on loans originated from 1999 to 2013, validation on 2014–2016, and test on 2017 onward. Exact boundaries should be adjusted after inspecting event rates per vintage. The financial crisis period can be evaluated separately as a stress scenario.

8. Model Families and Baselines

Table 2 summarizes all model families included in the experimental study.

Table 2: Model families and baselines. LR = logistic regression; LGBM = LightGBM; Cox PH = Cox proportional hazards; Weibull AFT = Weibull accelerated failure time; RSF = random survival forest; GBSA = gradient-boosted survival (AFT/Cox objective); RRSM = proposed reliability-regularized survival model. Deep models are optional.

Model	Family	Censoring	Competing Risks	Library
LR (12/24/36m)	Classification	No	No	scikit-learn
LGBM (12/24/36m)	Classification	No	No	LightGBM
Cox PH	Survival	Yes	No	lifelines
Weibull AFT	Survival	Yes	No	lifelines
RSF	Survival	Yes	No	scikit-survival
GBSA	Survival	Yes	No	XGBoost/LGBM
Cause-specific PH	Competing Risks	Yes	Yes	lifelines
Fine-Gray	Competing Risks	Yes	Yes	lifelines / R
DeepSurv	Deep Survival	Yes	No	PyTorch (optional)
DeepHit	Deep Survival	Yes	Yes	PyTorch (optional)
RRSM (proposed)	Survival + Reg.	Yes	No	Custom (PyTorch/XGB)

9. Reliability-Aware Survival Model (RRSM)

9.1. Model Structure

The proposed model is a gradient-boosted survival model augmented with a cohort-level hazard-ordering regularization term. The base model is trained using a Cox partial

log-likelihood or an AFT log-likelihood objective. Let $\hat{h}(\cdot \mid \mathbf{x}_i)$ denote the predicted hazard function for loan i with feature vector $\mathbf{x}_i$.

9.2. Cohort Pair Definitions

Cohort pairs $\mathcal{C} = \{(A_k, B_k)\}_{k=1}^K$ are defined based on origination characteristics, where cohort A_k is expected to have higher hazard than cohort B_k . Table 3 lists the pairs used in this study.

Table 3: Cohort pairs for hazard-ordering regularization. The “expected direction” is based on standard credit-risk conventions and serves as a modeling prior, not an empirical guarantee.

Cohort A (higher expected hazard)	Cohort B (lower expected hazard)	Feature basis
FICO < 620, LTV > 90%	FICO > 720, LTV < 70%	FICO + LTV
DTI > 45%	DTI < 36%	Debt-to-income
90+ days delinquent	Current	Current status
Investor-occupied	Owner-occupied	Occupancy type
First-time buyer	Repeat buyer	Buyer type
High-LTV (LTV > 90%)	Low-LTV (LTV < 80%)	LTV only

9.3. Training Objective

The total training objective is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{surv}} + \lambda \mathcal{L}_{\text{reg}}, \quad (8)$$

where $\mathcal{L}_{\text{surv}}$ is the base survival loss and the regularization term is:

$$\mathcal{L}_{\text{reg}} = \frac{1}{|\mathcal{C}|} \sum_{(A,B) \in \mathcal{C}} \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \max(0, \bar{h}_B(t) - \bar{h}_A(t)), \quad (9)$$

with $\bar{h}_A(t)$ and $\bar{h}_B(t)$ denoting the mean predicted hazard at time t over loans in cohort A and B respectively. The hyperparameter λ is tuned on the validation set using calibration error as the criterion.

9.4. Hazard-Order Violation Rate

The hazard-order violation rate (HOVR) for a cohort pair (A, B) is:

$$\text{HOVR}(A, B) = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \mathbf{1}(\bar{h}_B(t) > \bar{h}_A(t)). \quad (10)$$

The aggregate HOVR is averaged across all cohort pairs in $\mathcal{C}$. A lower HOVR indicates that the model’s predictions are more consistent with economic priors.

9.5. Mean Residual Lifetime Estimation

For loans active at the test cutoff date, estimated mean residual lifetime is computed from the fitted survival curve:

$$\hat{m}(t_0) = \frac{\int_{t_0}^{T_{\max}} \hat{S}(u) du}{\hat{S}(t_0)}, \quad (11)$$

approximated numerically over a finite horizon $T_{\max}$.

10. Experimental Plan

Ten experiments are planned in sequence. Table 4 gives an overview. The experiments are described briefly below; fuller implementation details are in the system development plan.

Experiment 1: Classification vs. survival modeling. Logistic regression and LightGBM classifiers are trained at fixed 12, 24, and 36-month horizons and compared against survival models on C-index, time-dependent AUC, and Integrated Brier Score. The hypothesis is that survival models perform better at longer horizons where censoring is more prevalent.

Experiment 2: Regularization ablation. Three variants are compared: RRSN with ordering regularization; the same model with $\lambda = 0$ (no regularization); and a sanity-check version with randomly assigned cohort orderings. If the regularization is useful, the ordered variant should have lower HOVR and better calibration than the unconstrained one; the random-ordering variant should perform worse than both.

Experiment 3: Random vs. time-based split. Results from random train-test splits are compared against vintage-based splits. The expected finding—that random splits overestimate deployment performance—is well-established in the literature, but documenting it on this specific dataset is important for interpreting all other results.

Experiment 4: Calibration. Raw model output is compared against post-hoc calibration via Platt scaling and isotonic regression at each prediction horizon. Calibration plots and expected calibration error are the primary evaluation tools.

Experiment 5: Feature set ablation. Four configurations are compared: origination features only; origination plus current loan state; origination plus rolling delinquency history; and the full time-varying panel. Performance is expected to improve with more longitudinal information.

Experiment 6: Competing risks. A default-only model (treating prepayment as censored) is compared against cause-specific and Fine–Gray subdistribution hazard formulations. This experiment evaluates how much the competing-risks treatment affects default probability estimates.

Experiment 7: Cohort hazard-ordering analysis. For each of the six cohort pairs in Table 3, the model’s predicted cohort-level hazard curves are compared against Kaplan–Meier estimates. HOVR is computed as the fraction of time points at which the predicted ordering contradicts the expected economic ordering.

Experiment 8: Mean residual lifetime. For loans active at the test cutoff, estimated MRL is computed from the fitted survival curve and compared across cohorts. Since ground truth is unavailable for active loans, this experiment evaluates internal consistency and interpretability rather than predictive accuracy.

Experiment 9: Statistical robustness. All main experiments are repeated across five random seeds with 100 bootstrap samples. Results are reported as means with 95% confidence intervals. Comparisons between models are considered meaningful only if confidence intervals do not substantially overlap.

Experiment 10: Portfolio-level evaluation. Five portfolio metrics are computed: expected default count error, top-decile and top-5% default capture rates, risk-band migration, and the approval-rate versus risk tradeoff. These metrics translate model performance into terms relevant to a credit risk team.

11. Experiment Matrix

Table 4: Full experiment matrix. HOVR = hazard-order violation rate; IBS = Integrated Brier Score; ECE = expected calibration error; MRL = mean residual lifetime.

Exp.	Comparison	Models	Primary Metrics
1	Classification vs. survival	LR, LGBM, Cox, AFT, RSF, GBSA, RRSM	C-index, AUC, IBS
2	Regularization ablation	RRSM (none/ordered/random)	HOVR, ECE, IBS
3	Split protocol	Random / vintage / stress	AUC, C-index, drop
4	Calibration method	None / Platt / isotonic	ECE, slope
5	Feature set	Origination / +state / +history / full	C-index, IBS
6	Competing risks	Default-only / prepay / CS-PH / Fine-Gray	Cause C-index
7	Cohort ordering	6 cohort pairs; RRSM vs. Cox	HOVR, survival curves
8	MRL analysis	All models on active loans	MRL comparison
9	Statistical robustness	All main models, 5 seeds, bootstrap	CI width, variance
10	Portfolio evaluation	RRSM vs. best baseline	EL error, top-decile

12. Evaluation Metrics

Table 5: Evaluation metrics. $\uparrow$ = higher is better; $\downarrow$ = lower is better.

Metric		Definition / Interpretation	Dir.	Experiments
C-index		Concordance index; fraction of correctly ordered risk pairs. Random = 0.5.	$\uparrow$	1, 2, 5, 6, 9
Time-dep. AUC (12/24/36m)		AUC for time-dependent case-control pairs at each horizon [Blanche et al., 2013].	$\uparrow$	1, 3, 5, 6
PR-AUC		Precision-recall AUC; informative under class imbalance.	$\uparrow$	1, 5, 9
Integrated Score	Brier	Weighted average Brier score over time; measures calibration and sharpness jointly [Graf et al., 1999].	$\downarrow$	1, 2, 5
Brier Score (fixed horizon)		Mean squared error of predicted probability vs. binary event indicator at a fixed horizon.	$\downarrow$	1, 4
Calibration slope		Slope of observed events on log-odds of predicted probability; ideal = 1.0.	near 1	4, 8
Expected calibration error (ECE)		Mean absolute difference between predicted probability and observed event rate per bin.	$\downarrow$	2, 4, 8
HOVR		Fraction of cohort-time pairs with predicted ordering violation.	$\downarrow$	2, 7
MRL comparison		Mean estimated remaining lifetime by cohort; evaluated for consistency.	—	8
Top-decile default capture	default	Fraction of defaults in the top 10% highest-risk loans.	$\uparrow$	1, 10
Top-5% default capture	default	Fraction of defaults in the top 5% highest-risk loans.	$\uparrow$	1, 10
Expected loss error		Absolute difference between predicted and observed default count at portfolio level.	$\downarrow$	10
Out-of-time drop		Difference in primary metric between in-time and out-of-time test sets.	$\downarrow$	3, 9

13. Calibration and Model-Risk Analysis

Calibration is the property that a model’s predicted probabilities agree with observed event frequencies. A model that assigns a default probability of 0.2 to a set of loans

should, on average, see approximately 20% of those loans default over the relevant horizon. Poor calibration can arise from systematic over- or under-prediction, miscalibration at extreme probabilities, or drift between training and deployment distributions.

For survival models, calibration should be assessed separately at each prediction horizon. A calibration plot at horizon τ bins loans by predicted $\hat{S}(\tau)$ and compares the mean predicted survival probability against the Kaplan–Meier estimate in each bin. Departure from the 45-degree diagonal indicates miscalibration.

Post-hoc calibration techniques, including Platt scaling and isotonic regression, can correct systematic miscalibration. They introduce additional model components that must be fit on held-out data and may not transfer across vintages if the calibration relationship shifts over time.

Model-risk considerations extend beyond calibration. A credit-risk model deployed in an automated approval or pricing pipeline carries quantifiable financial risk if it is miscalibrated, if it degrades under distribution shift, or if it is used outside the population on which it was validated. Time-based validation, stress-period evaluation, and explicit measurement of calibration drift between the validation and test sets are the main tools for assessing these risks in this study.

14. Ablation Studies

Table 6 specifies the ablation study design. Each row corresponds to a model variant in which one component is removed or modified relative to the full proposed model.

Table 6: Ablation study plan. Each variant is compared against the full RRSM on the time-based test set with all other hyperparameters held fixed. Results are reported with bootstrap 95% confidence intervals.

Variant	Removed / Changed	Purpose	Key Metric
RRSM (full)	— (reference)	Reference model	All
No regularization	$\lambda = 0$	Tests ordering value	HOVR, ECE
Random cohort ordering	Orderings reversed randomly	Sanity check	HOVR, IBS
No calibration	Skip post-hoc step	Tests calibration	ECE, slope
Origination only	Remove time-varying features	Tests feature value	C-index, IBS
No mixture component	Single-population model	Tests heterogeneity	C-index, ECE
Random split	Replace vintage split	Tests split bias	AUC, drop
Classification baseline	Replace with LR/LGBM	Tests survival frame	AUC, IBS

15. Robustness and Statistical Uncertainty

A common weakness in applied ML papers is reporting only point estimates without confidence intervals. When models differ by small margins, it is impossible to draw reliable conclusions without quantifying uncertainty. This project addresses robustness through two mechanisms.

Multiple random seeds. All models that involve random initialization are trained across five random seeds. The reported metric is the mean and standard deviation across seeds. A model variant whose performance is highly variable across seeds is considered less reliable, independent of its mean performance.

Bootstrap confidence intervals. For each model, 100 bootstrap samples are drawn from the test set. The metric of interest is computed on each bootstrap sample, and the 2.5th and 97.5th percentiles are reported as the 95% confidence interval. Comparisons between models are considered meaningful only if confidence intervals do not substantially overlap.

Out-of-time degradation. For each model, the primary metric is computed on both the validation set (in-time) and the test set (out-of-time). The out-of-time drop is defined as the difference between these two values. A plausible hypothesis is that reliability-regularized predictions, being partially governed by economic constraints that are more stable across vintages than arbitrary data patterns, may show smaller out-of-time degradation than unconstrained models. This remains to be tested.

16. Portfolio-Level Fintech Evaluation

Individual loan-level metrics measure discrimination and calibration but do not directly answer the question a risk team faces: if we use this model to rank loans and act on the top-risk segment, how many actual defaults do we capture?

Expected loss estimation error. For a portfolio of n active loans, the expected default count is $\hat{N}(\tau) = \sum_{i=1}^n \hat{p}_i(\tau)$. The estimation error is $|\hat{N}(\tau) - N_{\text{obs}}(\tau)|$, measuring suitability for provisioning and capital allocation.

Top- $k\%$ default capture. Loans are ranked by predicted default probability. The top- $k\%$ capture rate is the fraction of all observed defaults that fall in the top $k\%$ of the risk ranking. For $k = 10$, a random model captures roughly 10% of defaults; a useful model should capture substantially more.

Risk-band migration. Loans are assigned to risk bands (low, medium, high) based on predicted probability quantiles at origination. The fraction of loans transitioning between bands over time provides a measure of model stability and usefulness for portfolio monitoring.

Approval-rate versus risk tradeoff. For a given risk threshold, the approval rate is the fraction of loans with predicted risk below the threshold, and the associated risk is the expected default rate among approved loans. This tradeoff curve communicates model value to credit policy teams.

17. Figures and Visualization Plan

This section organizes the planned figures into three staged groups corresponding to the progression of the research: problem setup and survival background (Stage I), model evaluation and reliability-aware experiments (Stage II), and robustness, portfolio value, and explainability (Stage III).

17.1. Stage I: Problem Setup and Core Survival Concepts

The following six figures explain the problem formulation, survival analysis background, and dataset construction. All are conceptual and do not require experimental results; TikZ source is provided directly.

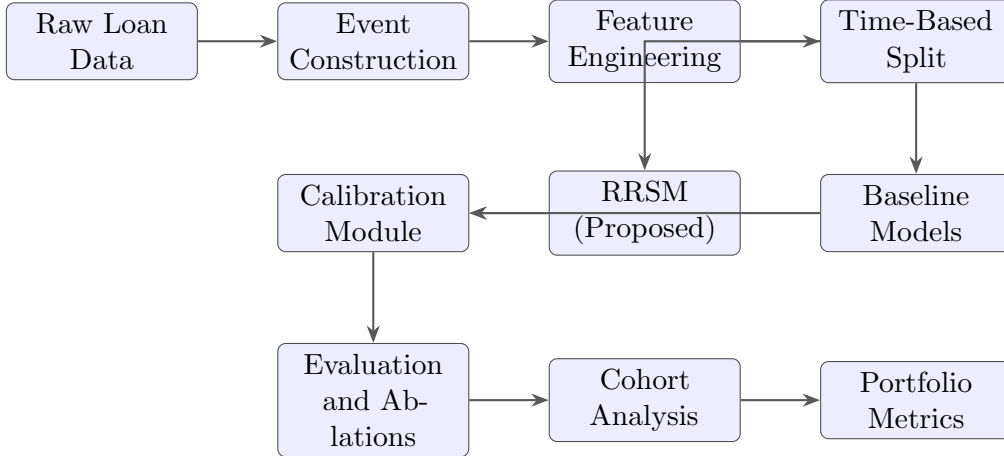


Figure 1: Research pipeline from raw loan data to portfolio-level evaluation. RRSM = Reliability-Regularized Survival Model.

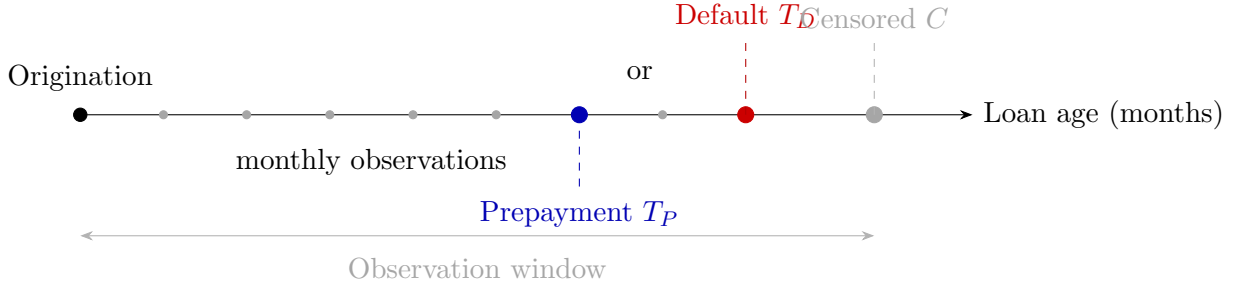


Figure 2: Loan lifecycle. A loan terminates as a default (red) or prepayment (blue), or is right-censored at the data cutoff (gray). The two terminal events are mutually exclusive.

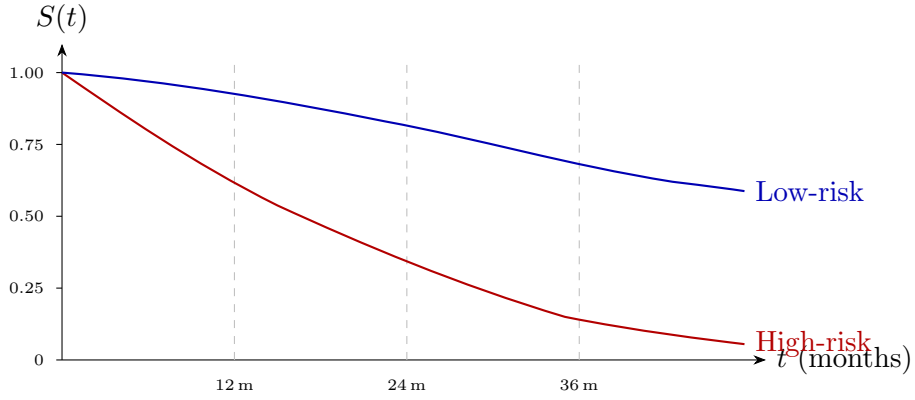


Figure 3: Illustrative survival functions $S(t)$ for high-risk (red) and low-risk (blue) borrower groups at 12, 24, and 36-month horizons. Values are schematic only.

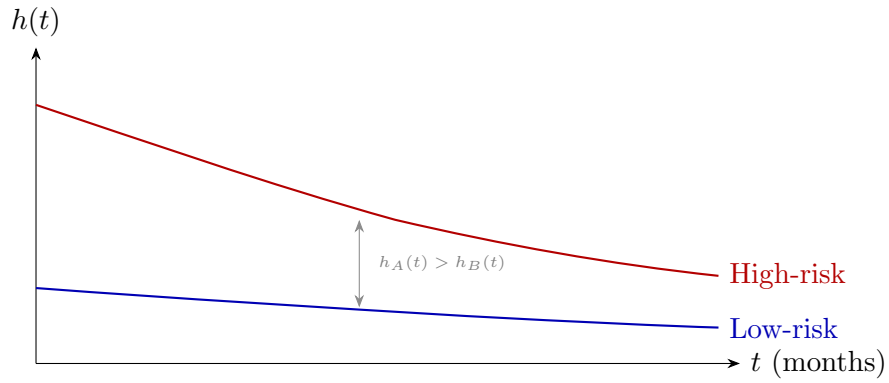


Figure 4: Illustrative hazard functions $h(t)$ for high-risk cohort A (red) and low-risk cohort B (blue). The RRSN penalizes violations of the ordering $h_A(t) > h_B(t)$ at every time point. Values are schematic only.

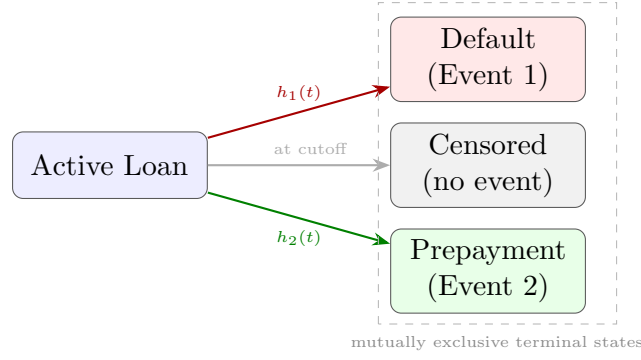


Figure 5: Competing-risks structure. An active loan transitions to default or prepayment through cause-specific hazards $h_1(t)$ and $h_2(t)$, or is censored at the data cutoff.

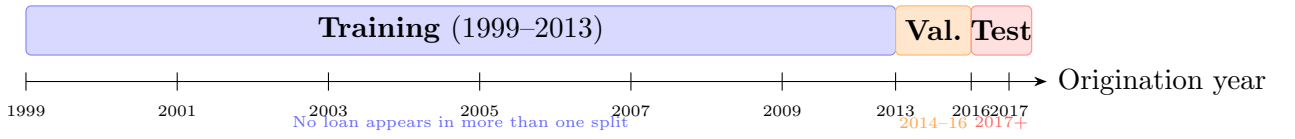


Figure 6: Vintage-based train / validation / test split. Exact year boundaries are subject to adjustment after inspecting per-vintage event rates.

17.2. Stage II: Model Evaluation and Reliability-Aware Experiments

Figures in this stage require numerical results from the corresponding experiments; compact placeholders are shown below and will be replaced once those experiments are completed.

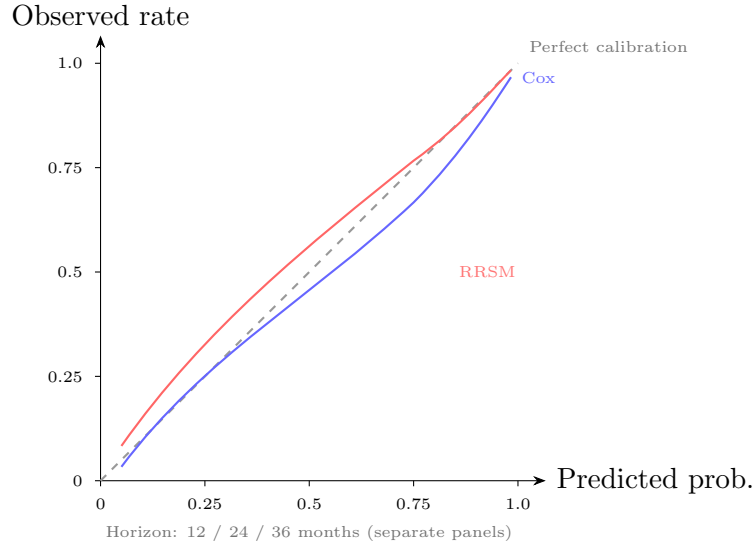


Figure 7: Calibration plot structure: predicted vs. observed default probability by decile for each model and horizon. Curves closer to the diagonal indicate better calibration. Numerical values pending Experiment 4.

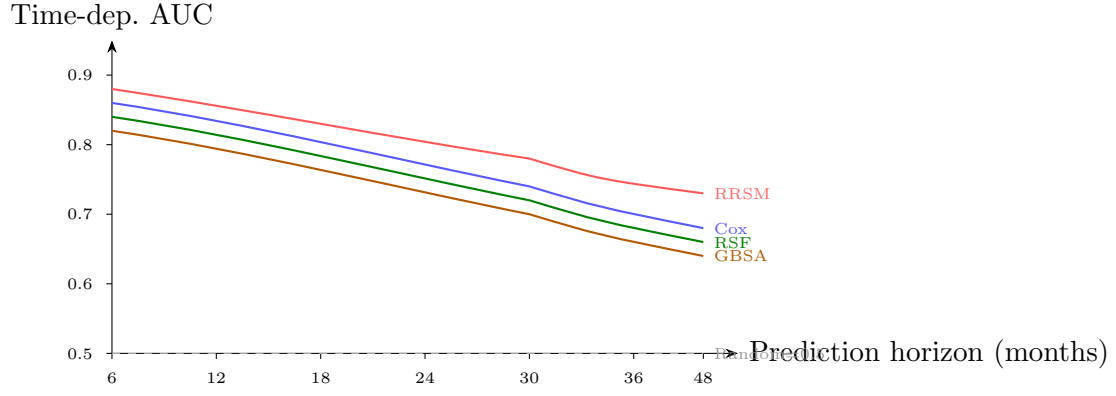


Figure 8: Time-dependent AUC over prediction horizon for all model families. Curve shapes are schematic; numerical values pending Experiment 1.

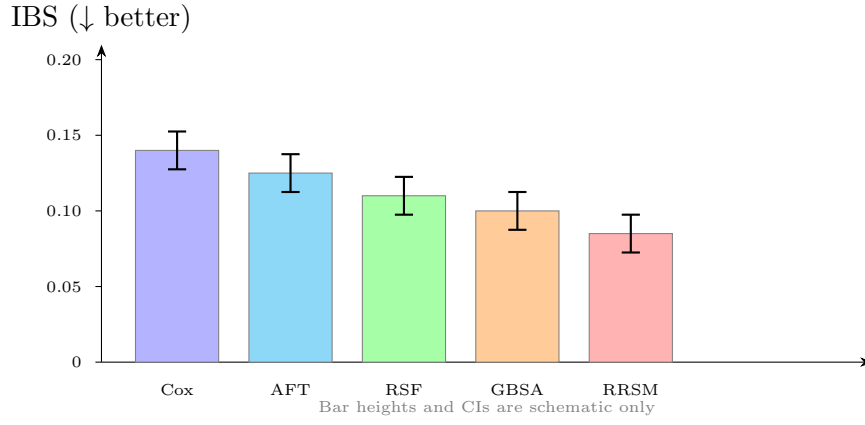


Figure 9: Integrated Brier Score by model with 95% bootstrap confidence intervals. Bar heights and error bars are schematic; numerical values pending Experiment 1.

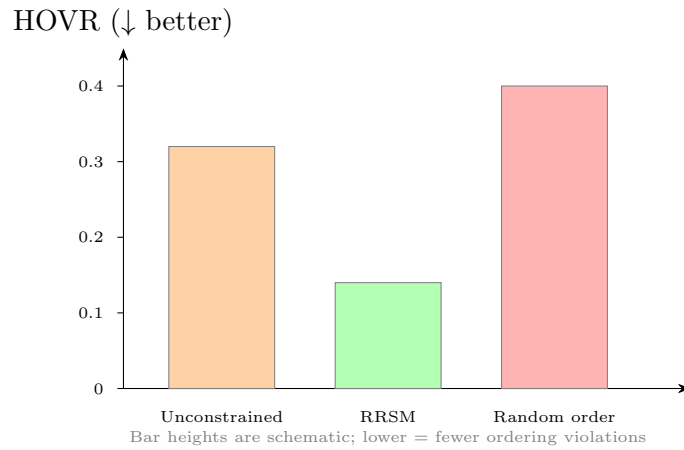


Figure 10: Hazard-order violation rate (HOVR) for unconstrained model, RRSM, and random-ordering sanity baseline. Bar heights schematic; numerical values pending Experiment 2.

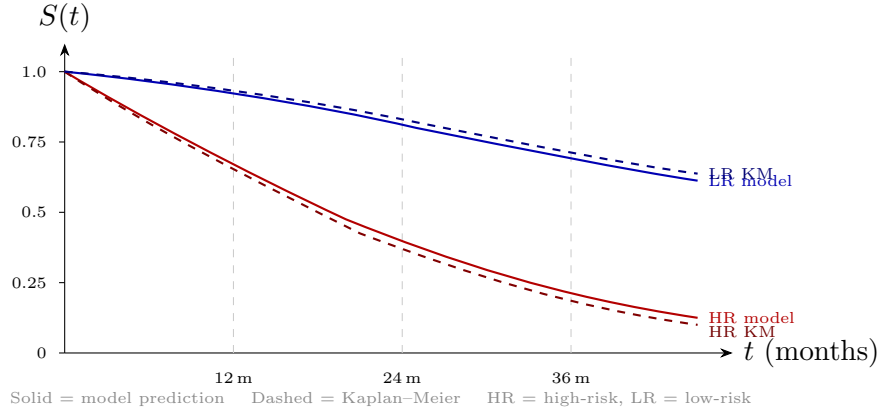


Figure 11: Predicted vs. Kaplan–Meier survival curves for high-risk and low-risk cohorts. Curve shapes are schematic; numerical values pending Experiment 7.

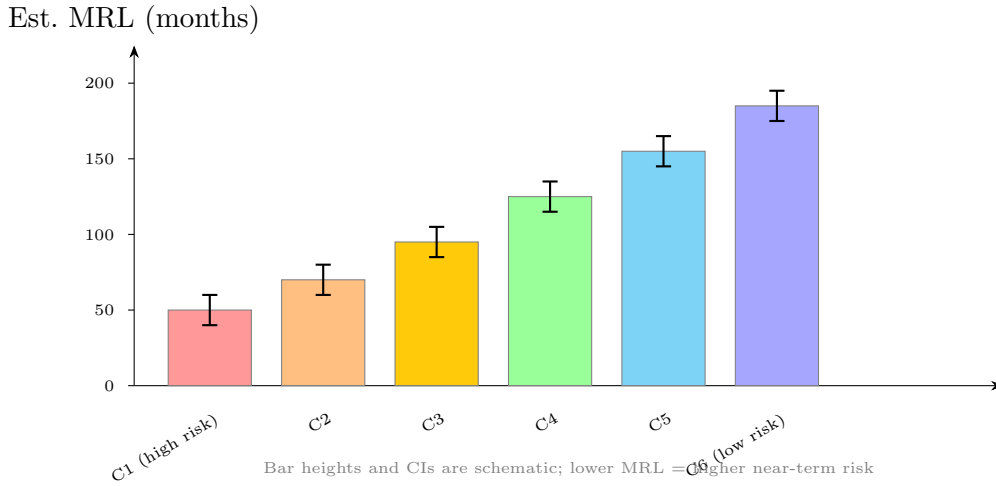


Figure 12: Estimated mean residual lifetime (months) by borrower cohort for active loans at the test cutoff. Bar heights schematic; numerical values pending Experiment 8.

17.3. Stage III: Robustness, Portfolio Value, and Explainability

Figures in this stage require numerical results from the robustness and portfolio experiments; compact placeholders are shown below and will be replaced once those experiments are completed.

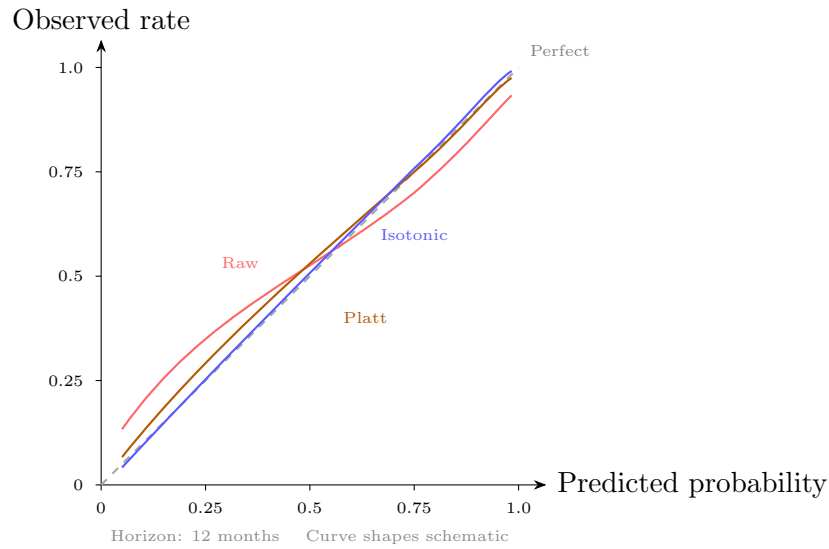


Figure 13: Calibration curves before and after post-hoc correction at the 12-month horizon. Curve shapes are schematic; numerical values pending Experiment 4.

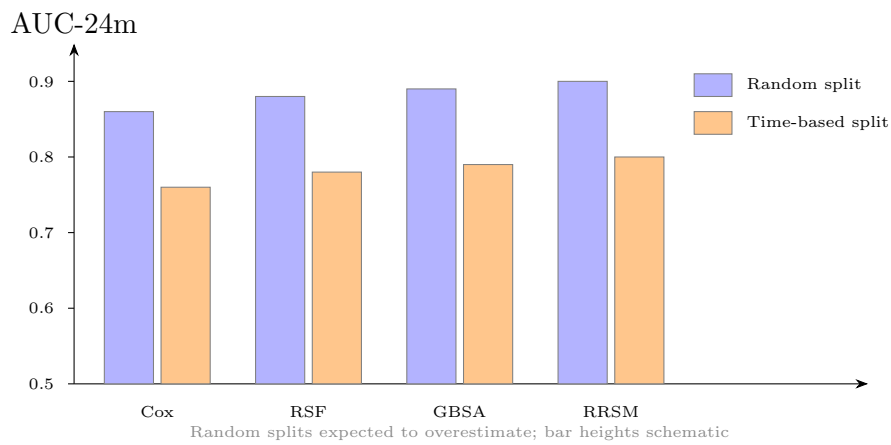


Figure 14: AUC-24m under random split vs. time-based vintage split by model. Random splits typically overestimate deployment performance. Bar heights schematic; numerical values pending Experiment 3.

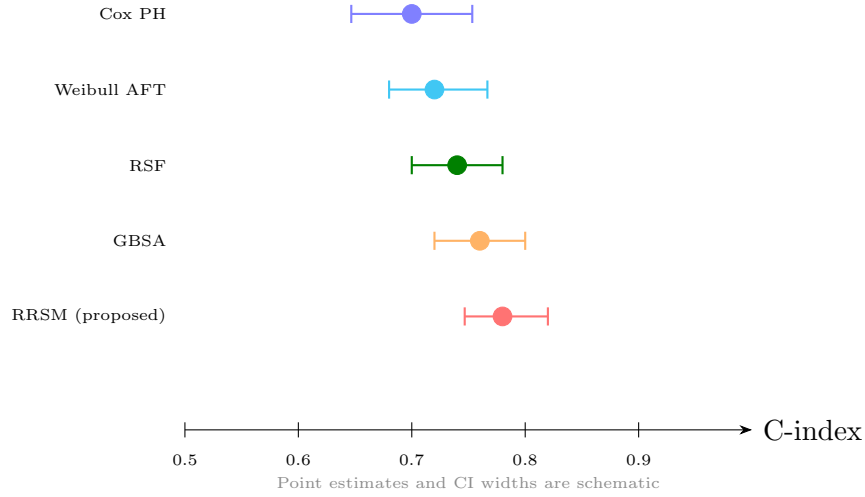


Figure 15: C-index with 95% bootstrap confidence intervals for all main models. Point positions and CI widths are schematic; numerical values pending Experiment 9.

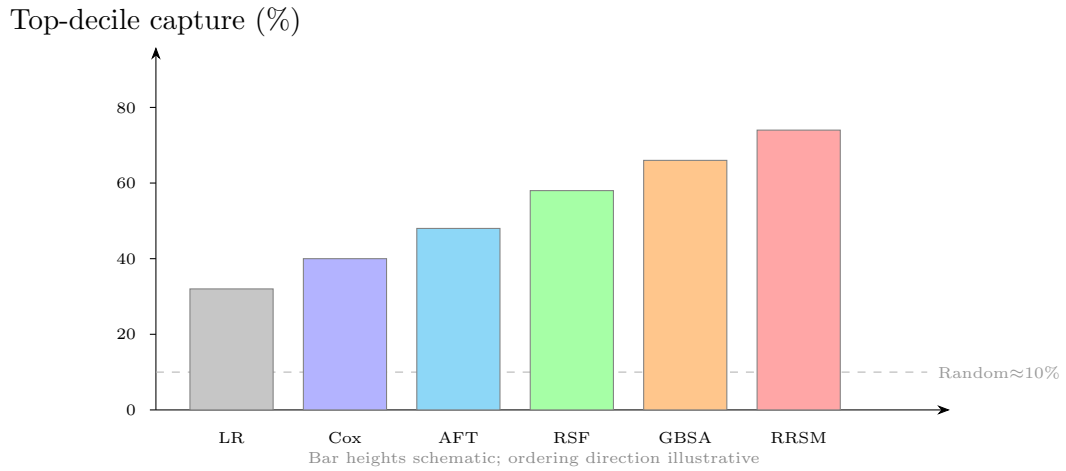


Figure 16: Top-decile default capture rate (fraction of observed defaults in the top 10% of risk-ranked loans) by model. Bar heights schematic; numerical values pending Experiment 10.

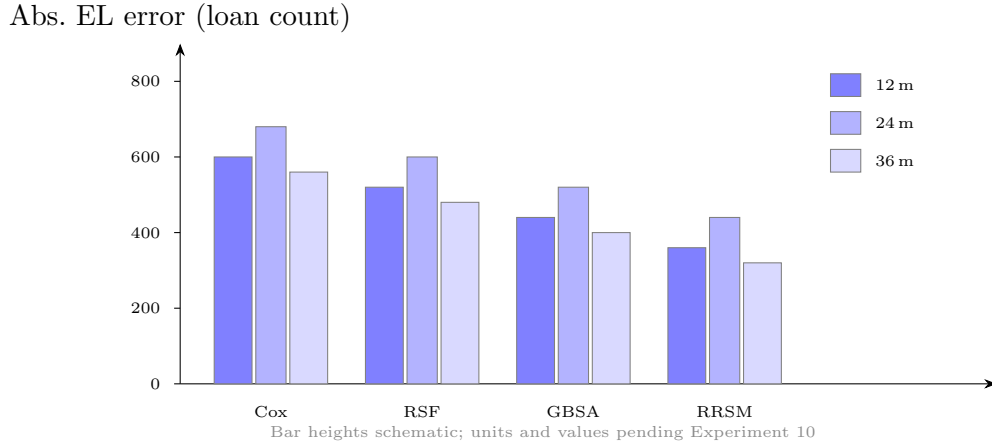


Figure 17: Expected loss estimation error (absolute difference between predicted and observed default count) at 12, 24, and 36 months by model. Bar heights schematic; numerical values pending Experiment 10.

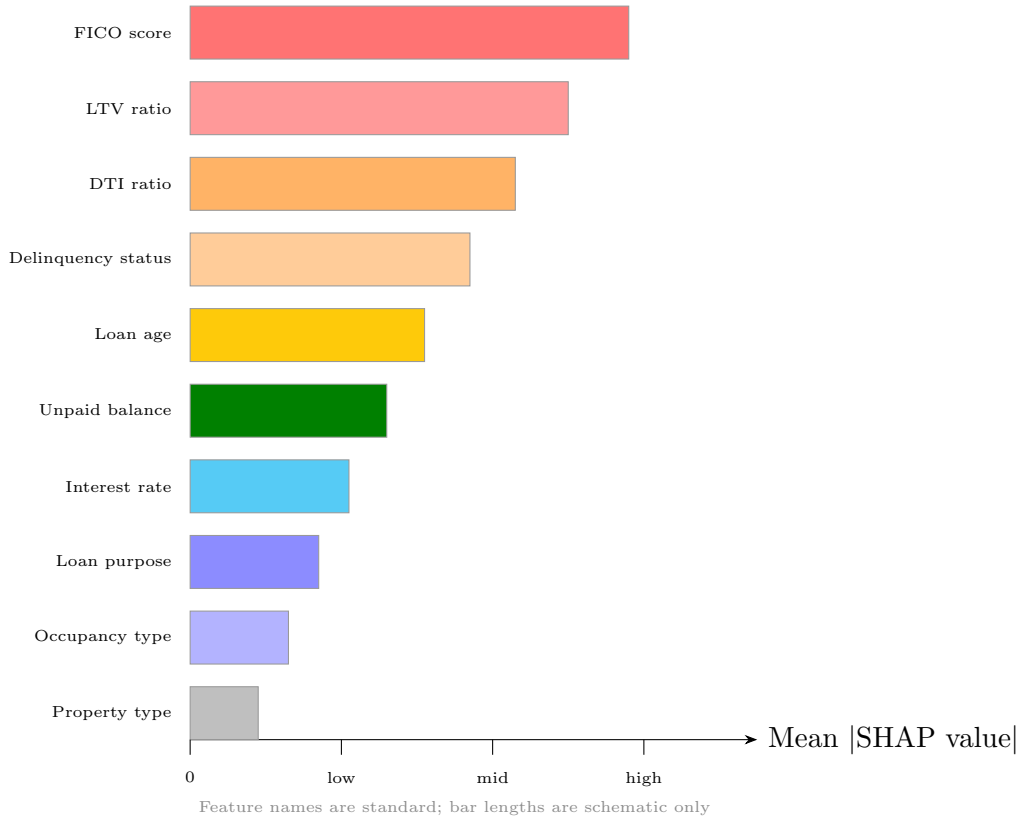


Figure 18: SHAP feature importance for 12-month default probability prediction (gradient-boosted model). Feature names are standard; bar lengths are schematic and numerical values are pending Experiment 10.

18. Expected Result Tables

Tables 7 and 8 are reporting templates. All numerical entries are pending experimental results.

Table 7: Main results: model comparison on the time-based test set. Values will be reported as means over five random seeds with 95% bootstrap confidence intervals in parentheses. IBS = Integrated Brier Score; ECE = expected calibration error at 12 months; HOVR = hazard-order violation rate. “—” = metric not applicable. **All numerical entries are pending experimental results.**

Model	C-index	AUC-24m	IBS	ECE-12m	HOVR
LR (24m)	Pending	Pending	—	Pending	—
LGBM (24m)	Pending	Pending	—	Pending	—
Cox PH	Pending	Pending	Pending	Pending	Pending
Weibull AFT	Pending	Pending	Pending	Pending	Pending
RSF	Pending	Pending	Pending	Pending	Pending
GBSA	Pending	Pending	Pending	Pending	Pending
RRSM (proposed)	Pending	Pending	Pending	Pending	Pending

Table 8: Ablation results on the time-based test set. RRSM (full) is the reference model. $\uparrow$ = higher is better; $\downarrow$ = lower is better. **All numerical entries are pending experimental results.**

Variant	C-index $\uparrow$	IBS $\downarrow$	ECE-12m $\downarrow$	HOVR $\downarrow$
RRSM (full)	Pending	Pending	Pending	Pending
No regularization	Pending	Pending	Pending	Pending
Random cohort ordering	Pending	Pending	Pending	Pending
No calibration	Pending	Pending	Pending	—
Origination features only	Pending	Pending	Pending	Pending
No mixture component	Pending	Pending	Pending	Pending
Random split (RRSM)	Pending	Pending	Pending	—
Classification baseline	Pending	—	Pending	—

19. Experimental Design for the Fintech Credit-Risk System

19.1. Feature Engineering

Static origination-level features include credit score, LTV, DTI, loan amount, interest rate, loan term, loan purpose, occupancy type, property type, and geographic region. Time-varying features include current delinquency status, current unpaid principal balance, loan age, remaining months to maturity, modification flag, and rolling delinquency count over the prior three and six months. Derived features relevant to survival modeling include vintage, loan seasoning, and cohort-level empirical hazard estimates.

19.2. Handling Class Imbalance

Default events are rare in most mortgage portfolios. Precision-recall analysis is more informative than ROC analysis under severe imbalance because PR curves are not affected by the large number of true negatives. Standard machine learning models, particularly boosted trees, are often miscalibrated under class imbalance and benefit from post-hoc calibration via isotonic regression or Platt scaling.

20. Limitations and Open Research Questions

Public mortgage data captures a specific segment of the credit market. Conclusions drawn from Freddie Mac and Fannie Mae data may not generalize directly to unsecured lending, BNPL, or credit-card portfolios, which have different loan structures, prepayment dynamics, and borrower demographics.

Temporal distribution shift is a genuine challenge. Mortgage default dynamics during the 2008–2010 financial crisis were qualitatively different from those in low-volatility periods, and a model trained on one regime may not be well-calibrated in another. Reliability-aware regularization may help by imposing constraints that are economically motivated rather than purely data-driven, but this remains to be tested empirically.

The choice of cohort definitions for hazard-ordering regularization is non-trivial. Poorly chosen cohorts may impose constraints that are not economically meaningful, and overly aggressive regularization may harm model flexibility. The weight λ on the ordering penalty requires careful tuning.

Several research questions remain open. How should the ordering regularization be specified when cohort hazard orderings are expected to be time-varying rather than uniformly monotone? Is there a principled Bayesian approach to incorporating hazard-ordering priors? How should the latent mixture model’s number of components be selected? Can reliability-aware survival models be adapted to panel data with time-varying covariates in a fully consistent way? These are reasonable directions for further work.

21. Threats to Validity

Internal validity. The main threat to internal validity is improper data leakage between train and test sets. Because the dataset is longitudinal, a naive row-level random split can place later observations from the same loan in both sets, or allow features computed from the full loan history to carry future information into training. The time-based vintage split addresses the first concern; careful feature construction (using only lookback windows anchored to the current observation month) addresses the second.

The choice of cohort pairs for hazard-ordering regularization introduces a prior that might be misspecified. If the economic ordering does not hold empirically for a specific origination period or macro environment, the regularization may impose incorrect constraints. This sensitivity should be tested by evaluating HOVR on empirical Kaplan–Meier hazard estimates separately from the model predictions.

External validity. Freddie Mac and Fannie Mae data represent conforming single-family mortgages. The borrower population, loan structures, and prepayment behavior differ from non-conforming, subprime, BNPL, or credit-card portfolios. Conclusions about model performance should be stated as applying to the specific dataset and not generalized to broader lending contexts without further validation.

Construct validity. Event definitions derived from dataset status codes are proxies for the economic concepts of default and prepayment. Different code-to-event mappings can produce different event rates and model rankings. The sensitivity of results to event definition choices should be assessed by testing alternative threshold definitions for default.

Statistical validity. Given the large sample sizes typical of mortgage datasets, even small and economically unimportant performance differences may reach statistical significance. Confidence intervals and effect sizes should be reported alongside p -values where significance testing is used. Model comparisons based on overlapping confidence intervals are more conservative and more meaningful than point-estimate comparisons.

22. Conclusion

Binary classification is a natural starting point for credit default prediction but an incomplete one. Survival analysis provides a framework that handles censoring, models event timing rather than just event occurrence, and produces outputs—hazard curves, survival probabilities, remaining lifetime estimates—that are directly useful for credit risk management.

Reliability theory contributes complementary ideas: hazard ordering, stochastic ordering, and mixture models that formalize what we expect from borrower cohorts and provide principled ways to incorporate prior knowledge into estimation. Combining these ideas with flexible machine learning models is a natural extension that has begun to receive attention in the literature but remains underexplored in applied credit modeling.

A reliability-regularized survival ML framework, trained on large-scale public mortgage data with careful event definitions, time-based validation, and rigorous ablation, could contribute a useful benchmark for the field. The key empirical claims—that reliability-

aware regularization improves calibration, that hazard-ordering constraints reduce violation rates, and that survival models outperform classification models at medium-to-long horizons—are testable and will be evaluated honestly, including in cases where the proposed approach does not clearly win.

System Development Plan

Data Ingestion. Download the Freddie Mac Single-Family Loan-Level Dataset from the official research portal. The dataset consists of two flat files per origination year: an origination file with static loan characteristics and a monthly performance file with loan-level records for each active reporting month. Parse both files using Python (**pandas** or **polars**), ensuring correct handling of column delimiters, missing-value codes, and categorical codes as defined in the official user guide. Start with a subset of origination years to keep the development loop fast.

Preprocessing. Join origination and performance files on the loan identifier. Sort the resulting panel by loan and month. Check for duplicate records, missing credit scores, anomalous LTV values, and implausible loan ages. Document all filtering decisions. Normalize continuous features. Encode categorical variables using ordinal or one-hot encoding depending on cardinality. Create a clean loan-month panel with one row per active loan-month.

Event Definition. Using the data dictionary, define a terminal status column that takes one of three values: default, prepayment, or censored. Compute the time-to-event in months from the first observed month to the event month. Verify event rates against published statistics if available. Document all definitional choices.

Feature Engineering. Construct a static feature matrix from origination-file columns (credit score, LTV, DTI, loan amount, interest rate, term, purpose, property type, channel, state). Construct time-varying features from the performance file: current delinquency status, unpaid principal balance, loan age, remaining months to maturity, modification flag, and rolling delinquency count over the prior three and six months. Create cohort labels based on binned credit score and LTV for use in hazard-ordering evaluation.

Train/Validation/Test Split. Assign loans to split sets based on origination vintage. Suggested partition: training loans originated 1999–2013, validation 2014–2016, test 2017 onward. Confirm that no loan appears in more than one set. Report event rates, loan counts, and censoring rates in each split.

Baseline Models. Implement the following baselines in sequence: (1) logistic regression at 12-, 24-, and 36-month fixed horizons; (2) LightGBM binary classifier at the same horizons; (3) Cox proportional hazards model using **lifelines**; (4) Weibull AFT model; (5) random survival forest using **scikit-survival**. For each model, evaluate on the validation set and record the full metric set.

Survival ML Models. Implement gradient-boosted survival using the AFT or Cox objective in `XGBoost` or `LightGBM`. Tune main hyperparameters via time-based cross-validation. If computationally feasible, implement `DeepSurv` using `PyTorch`.

Reliability-Aware Regularization. Define cohort pairs based on origination characteristics. Implement the hazard-ordering penalty as a differentiable loss term. Tune λ using validation-set calibration error as the criterion. Track the HOVR as a secondary metric. Separately, implement MRL estimation for active loans by numerically integrating the predicted survival curve beyond the current loan age.

Evaluation Metrics. Compute on the test set: C-index, time-dependent AUC at 12, 24, and 36 months, IBS, calibration slope, ECE, PR-AUC, top-decile and top-5% default capture, HOVR, and MRL comparison. Report all metrics with bootstrap confidence intervals from at least 100 bootstrap samples. Repeat training with multiple random seeds.

Calibration. After model training, evaluate calibration on the validation set using calibration plots. Apply isotonic regression or Platt scaling if calibration is poor. Re-evaluate on the test set using the same protocol.

Explainability. Generate borrower-level survival curves for a representative sample of loans. Plot cohort-level mean survival and hazard curves for key cohort pairs. Compute SHAP values for fixed-horizon default probability predictions from the gradient-boosted model. Document whether predicted hazard orderings match economic expectations.

Ablation Study. Run all ablation variants in Table 6 and record the full metric set. Organize results in a comparison table with confidence intervals.

Reproducibility. Maintain a clean GitHub repository with a `README` describing the dataset access procedure, environment requirements, and step-by-step instructions for reproducing all results. Use configuration files to specify all hyperparameters. Fix random seeds and record them. Save all result tables and trained models as versioned outputs.

Optional Dashboard. A Streamlit dashboard can be developed as a secondary deliverable, displaying borrower-level survival curves and default probabilities at user-specified horizons, cohort comparison plots, top-risk loan segments, and portfolio-level expected loss summaries.

Implementation Timeline

Table 9: Suggested 12-week implementation timeline with deliverables.

Week(s)	Tasks	Deliverable
1–2	Literature review; dataset download; event-code mapping; preliminary EDA	Data dictionary n
3–4	Loan-month panel construction; feature engineering; vintage-based split	Processed dataset
5–6	Classification baselines (LR, LGBM); Cox and Weibull AFT	Baseline metric ta
6–7	RSF and GBSA survival models	Extended baseline
7–8	RRSM implementation; cohort pair definition; regularization tuning	RRSM training sc
8–9	Experiments 2–5 (ablation, split, calibration, feature sets)	Ablation table dra
9–10	Experiments 6–8 (competing risks, cohort ordering, MRL analysis)	Cohort analysis pl
10–11	Experiments 9–10 (robustness, portfolio metrics); figure generation	Full result tables;
11–12	Paper writing; repository cleanup; reproducibility check	Final draft; GitHu

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